

Motion compensation in the region of the coronary arteries based on partial angle reconstructions from short-scan CT data

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Purpose: In order to mitigate motion-induced artifacts, several motion compensation (MoCo) methods have been developed, which are either able to (a) compensate for severe artifacts, but utilize the data for the reconstruction of several cardiac phases, or (b) improve image quality of a single reconstruction with only moderate motion artifacts. We propose a method combining both benefits: dose efficiency by utilizing only the data needed for a single short-scan reconstruction while still being able to compensate for severe artifacts.

Methods: We introduce a MoCo method, which we call PAMoCo, to improve the visualization of the coronary arteries of a standard coronary CT angiography exam by reducing motion artifacts. As a first step, we segment a region of interest covering a chosen coronary artery. We subdivide a volume covering the whole heart into several stacks, which are sub-volumes, reconstructed from phase-correlated short-scan data acquired during different heart cycles. Motion-compensated reconstruction is performed for each stack separately, based on partial angle reconstructions, which are derived by dividing the data corresponding to the segmented stack volume into several double-overlapping sectors. We model motion along the coronary artery center line obtained from segmentation and the temporal dimension by a low-degree polynomial and create a dense 3D motion vector field (MVF). The parameters defining the MVF are estimated by optimizing an image artifact measuring cost function and we employ a semi-global optimization routine by re-initializing the optimization multiple times. The algorithm was evaluated on the basis of a phantom measurement and clinical data. For the phantom measurement an artificial vessel equipped with calcified lesions mounted on a moving robot arm was measured, where typical coronary artery motion patterns for 70 bpm and 90 bpm have been applied. For analysis, we calculated the calcified volume V inside an ROI and measured the maximum vessel diameter d based on cross-sectional views to compare the performances of standard reconstructions obtained via filtered backprojection (FBP) and PAMoCo reconstructions between 20% and 80% of the cardiac cycle. Further, the new algorithm was applied to six clinical cases of patients with heart rates between 50 bpm and 74 bpm. Standard FBP, PAMoCo reconstructions were performed and compared to best phase FBP reconstructions and another MoCo algorithm, which is based on motion artifact metrics (MAM), via visual inspection.

Results: In case of the phantom measurement we found a strong dependence of V and d on the cardiac phase in case of the FBP reconstructions. When applying PAMoCo, V and d became almost constant due to a better discrimination from calcium to vessel and water background and values close to the ground truth have been derived. In the clinical study we chose reconstructions showing strong motion artifacts and observed a substantially improved delineation of the coronary arteries in PAMoCo reconstructions compared to the standard FBP reconstructions and also the MAM images, confirming the findings of the phantom measurement.

Conclusions: Due to the fast reconstruction of PAMoCo images and the introduction of a new motion model, we were able to re-initialize the optimization routine at pre-selected parameter sets and thereby increase the potential of the MAM algorithm. From the phantom measurement we conclude that PAMoCo performed almost equally well in all cardiac phases and suggest applying the PAMoCo algorithm for single source systems in case of patients with high or irregular heart rates.

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Key words: cardiac CT, computed tomography (CT), motion artifacts, motion compensation, motion estimation

1. INTRODUCTION

In recent years, developments such as phase-correlated reconstruction,^{9,10} multislice detectors, multidetector systems, high-pitch scan modes, tube current modulation for spiral scans, low kV scanning, improvements in slice thickness, and rotation time, allowed coronary CTA to be used as an alternative to invasive coronary angiography. Cardiac CT imaging has rapidly become clinically relevant as it provides a noninvasive method for the diagnosis of coronary artery diseases via coronary CT angiography or being used as diagnostic postsurgical follow-up for the assessment of stent patency and the degree of a possible restenosis.^{8,21} Hence, several applications like calcium scoring or plaque component analysis have found their way into the clinical routine. In future, also the recently developed fractional flow reserve (FFR) CT measurements, which enable to determine the physiological significance of a stenosis in the presence of coronary artery disease (CAD), might find its way into the clinics.¹² However, trying to assess the degree of a stenosis, the severity of a calcification, or the lumen of a coronary artery, which has a diameter only in between 1 mm and 5 mm,¹³ places high demands on the spatial and temporal resolution of a reconstruction. Considering the temporal resolution t_{res} of a standard filtered backprojection reconstruction, which can be approximated for single-source CT (SSCT) systems by $t_{\text{res}} \approx t_{\text{rot}}/2$, with rotation time t_{rot} , close to the isocenter, displacements of up to 1 cm during data acquisition due to the heart beating may occur³ and might introduce a strong blurring in the region of the coronary arteries. As a consequence, so-called motion artifacts caused by an insufficient temporal resolution, are still the main cause of artifacts in cardiac CT.

The hardware-based solution to this problem was introduced in the form of dual-source CT (DSCT) systems,¹⁶ which are able to double the temporal resolution compared with a single-source CT system of identical rotation time. In case of a well chosen reconstruction phase, such a temporal resolution might be sufficient to freeze cardiac motion. However, until today, the utilization of SSCT systems is still widespread in cardiac CT exams.

In order to increase the temporal resolution of SSCT systems in cardiac CT, until today a plenty of software-based solutions have been developed. The algorithms can basically be divided into the group of iterative reconstruction algorithms,^{19,20,25} which utilize a data range ($180^\circ + \beta$, fan angle $\approx 50^\circ$, $\beta \approx 60^\circ$) that is smaller than the data range needed for a conventional reconstruction via filtered backprojection (FBP) ($180^\circ +$) to increase the temporal resolution of the latter. However, motion compensation (MoCo) algorithms, which increase the temporal resolution by estimating 3D or 4D motion vector fields (MVFs), form the majority of the proposed solutions, since they show a much greater

potential for the reduction of severe motion artifacts, as discussed by Hahn et al.⁴

1.A. Motion compensation in cardiac CT

There are several reasons to perform MoCo in cardiac CT. One motivation is to provide functional data at low patient dose but quasi-constant image quality over all cardiac phases.^{23,24,26} For these applications, usually 3D registration routines are employed, from which MVFs can be derived, enabling to warp each reconstruction of the functional dataset onto any other cardiac phase. However, the temporal resolution of such warped images cannot exceed the temporal resolution of the initial reconstruction. Another motivation for MoCo is to increase the image quality of a single cardiac phase, when no complete functional information is needed for diagnosis, for example, for coronary CTA with the aim of ruling out stenosis. The remaining methods in general differ in the utilized data range, the targeted region of interest (ROI), and the way motion is estimated and compensated for. In the following sections, we roughly classify the remaining MoCo algorithms based on the utilized data range into the following two basic groups:

1.A.1. Volume or landmark registration-based motion estimation

Volume registration-based methods or landmark registration-based methods utilize reconstructions of multiple, at least slightly shifted cardiac phases to determine a MVF correcting the motion in a single reconstruction of a targeted cardiac phase. While volume-based registration methods^{7,11} enable the compensation of motion in the complete targeted volume, landmark-based registration routines are more organ or vessel specific and enable to compensate motion in a targeted volume, which is also a way to reduce the computational effort. However, landmark-based registration routines depend on an initial segmentation of, for example, the coronary arteries^{2,6,15} or the heart surface,²⁸ whose position can be tracked in the utilized reconstructions. A combination of both versions, where motion estimation is split into a global part covering the heart and a local part, addressing the coronary arteries, was recently proposed by Tang et al.²⁷ The global registration also allows correcting artifacts in the region of the myocardium, which are also addressed by Nett et al.¹⁴ Furthermore, for a modified version of a volume registration-driven algorithm, which registers only two, incomplete so-called conjugate images, which are partial angle reconstructions from projection views separated by 180° , was proposed by Kim et al.¹¹

1.A.2. Motion artifact metric-based motion estimation

Another possibility to estimate motion, using only the data needed for the reconstruction of a single cardiac phase ($180^\circ +$) — the short-scan data — was proposed by Rohkohl et al.¹⁷ and was discussed by Hahn et al.³ In order to improve the image quality of a single reconstruction, motion artifact metrics (MAMs), for example, the image entropy, are used as a cost function whose minimization seeks for sharp, motion artifact-free images. Parameters defining the MVF, which describe a nonrigid motion of the vessel, can be optimized using a gradient descent routine, as proposed by Rohkohl et al.¹⁷ This way, the method is able to improve the image quality of images featuring slight motion artifacts. However, break-ins of image quality at certain cardiac phases have been observed, which are caused by the optimization being trapped in a local minimum caused by the nonconvexity of the cost function. This limits the scope of applications for this method to initial reconstructions not too strongly corrupted by motion artifacts, or respectively to improving the image quality of the best-phase reconstruction, which is the reconstruction of the cardiac phase featuring least motion artifacts. In the method presented by Hahn et al.,³ which also uses MAMs to estimate a MVF, an approach to overcome the issue of the nonconvex cost-function was proposed, which is based on reinitializing the optimization multiple times. However, although the temporal behavior of the MVF was modeled with a polynomial allowing to correct more than linear motion, a spatial variation was not taken into account, limiting the method to be applied only to small subvolumes.

Registration-based methods have the potential to compensate for strong motion artifacts. However, the data range used for motion estimation is in general at least extended to $180^\circ + \beta$, $\beta \approx 50^\circ$, since for the MoCo reconstruction of a single cardiac phase, several images at slightly deviating time points need to be generated. Furthermore, in general a spatially dense sampled MVF is created, whose accuracy depends on the accuracy of the registration. Hence, a reference phase of already quite good quality might be needed. In contrast to the spatial resolution of the MVF, the temporal resolution is defined by linear interpolation between at least three time points.

We propose an algorithm combining benefits from both groups A and B, since it is dose efficient, as it is based on short-scan data range, but still able to compensate for strong motion artifacts. However, to enable a fast computation with a limited set of parameters, we limit our method to an ROI covering the coronary arteries.

2. MATERIALS AND METHODS

2.A. Proposed method

In order to increase the temporal resolution in cardiac CT we propose a MoCo algorithm, which we call the partial angle-based motion compensation (PAMoCo). With regard

to dose minimization, we perform motion estimation and compensation from nothing more than the data needed for a short-scan reconstruction. For the MoCo, we focus on the region of the coronary arteries, which are highly relevant for diagnosis of cardiovascular diseases, stent evaluation or plaque quantification, where — when contrast is enhanced — motion artifacts are most strongly visible. Most of the methods from group A and B employ a computationally expensive, data-based motion compensation technique¹⁸ of incorporating the estimated motion during the backprojection step, by backprojecting along new trajectories. However, we adopt a potentially faster way to perform motion compensation, which was first suggested by Bhagalia et al.,² who called it the subphasic warp and add scheme, and which was later adopted by Hahn et al.³ According to this method, fast motion compensation is obtained, applying the MVFs directly on several partial angle reconstructions (PARs), which are incomplete reconstructions from only a subset of the short-scan data, and motion compensation is obtained by adding the warped PARs.

The algorithm consists of the following seven components: (a) initial reconstruction and coronary vessel segmentation, (b) PAR reconstruction, (c) motion compensation, (d) motion model, (e) motion vector field, and (f) motion estimation. Steps (b) to (f) are applied independently on several volumes overlapping in z -direction. The size of the so called stack volumes, which altogether cover the whole heart, is proportional to the size of the detector. The stacks are reconstructed from heart phase-correlated data associated with a certain cardiac phase c acquired during a single heart beat. As a consequence in cases of patients with irregular heartbeats or nonoptimal chosen gating positions, adjacent stacks might not correspond to the exactly same heart phase and we expect shifts between them. For this reason, our algorithm addresses the stacks separately and MoCo is performed independently. We will replace the MoCo reconstructions of the separate stacks in the original reconstruction in an adjoining, nonoverlapping manner leading to possible discontinuities at the borders, which are, however, not further addressed in this work.

2.A.1. Segmentation

In order to limit the computational effort of the method, we intend to compensate for motion only in the region of the coronary arteries, where the influence of motion artifacts on diagnosis is strongest. To perform a segmentation of a chosen coronary artery, we create an initial reconstruction f_c^0 from short-scan data correlated with heart phase c , such that for each individual image only data stemming from the chosen heart phase and a single heartbeat is used for the reconstruction of the volume, covering the whole heart. The temporal resolution of such an initial reconstruction $t_{\text{res}} \approx t_{\text{rot}}/2$, can be approximated by half of the rotation time t_{rot} ($t_{\text{res}} \approx 143$ ms in our case), assuming that the heart is close to the isocenter of the scan. In terms of scanning technique, the data can be acquired in retrospective spiral scan mode

with tube current modulation correlated with an ECG signal, or in a prospectively ECG-triggered sequence scan.

Based on the original short-scan reconstruction f_c^0 , of a chosen cardiac phase c , we perform an automated, prior knowledge-based segmentation of a chosen coronary artery by an algorithm, which returns a coronary center line $r_{CL}(\lambda)$. We herein parametrize the center line with the parameter $\lambda \in \mathbb{R}$, with $r_{CL}(\lambda) : \mathbb{R} \rightarrow \mathbb{R}^3$, being a bijective, continuous mapping of the center line coordinate to a voxel position r_{CL} on the center line. For the segmentation, we use an external software, which is also used in Syngo®.via (SIEMENS Healthcare GmbH) cardiac applications, which can distinguish between the left anterior descending and right coronary artery, as well as the left circumflex artery (CL = LAD, RCA, CX). MoCo will be performed for each branch separately.

We define the distance $d(r, \lambda) := |r - r_{CL}(\lambda)|$ of a voxel r to the center line at center line coordinate $\lambda \in [-1, 1]$ with central position $\lambda_0 = 0$ passing through the stack volume and generate a tubular ROI Ω_s of radius $2R$ around the chosen coronary artery for each stack separately

$$\Omega_s = \{r \in \mathbb{R}^3 | \forall \lambda \in [-1, 1] : d(\lambda) < 2R\}, \quad (1)$$

covering all voxels, with a distance closer than $2R$ to the center line.

We estimated R with the help of the coronary artery velocity measurement, where we assume a constant velocity during the scan, such that $R = v_{\max} \cdot t_{\text{rot}}/2 \approx 10$ mm, with $v_{\max} \approx 70$ mm/s being the maximum expected displacement of an artery during data acquisition and $t_{\text{rot}} = 285$ ms. This way we ensure that the ROI includes all motion artifacts relevant for compensation. The factor 2 in front of R in Eq. (1) allows for a smooth fading of the MVF to zero at the borders of Ω_s , which is described in Section II.A.5.

2.A.2. PAR reconstruction

The basis of the following MoCo is the decomposition of the images into PARs. In order to create those PARs, we create projection data $p'_c = Xf_c(\Omega_s)$ covering an angular range of 180° by forward projection on the segmented volume $f_c(\Omega_s)$. We divide these synthetically generated data p'_c , which we obtain in parallel beam geometry parametrized by the projection angle ϑ and the detector coordinate ξ , into $2K + 1$ double-overlapping segments, which are centered around the angles $\vartheta_k = \vartheta_0 + k\Delta\vartheta$ with $-K \leq k \leq K$ in angular increments of $\Delta\vartheta = \pi/(2K + 1)$. The segments are defined by introducing a weighting function $\omega_k(\vartheta)$ displayed in Fig. 1, which might, for example, be modeled by a $\cos^2$ -function. The PARs f_k are then obtained by FBP of the data segments $\omega_k(\vartheta)p'_c$:

$$f_k = X^{-1} \omega_k(\vartheta) p'_c.$$

Since we use synthetic data p'_c generated by forward projection of the segmented volume for the reconstruction of the PARs, the correlation between projection angle and acquisition time is not completely preserved, as is illustrated in

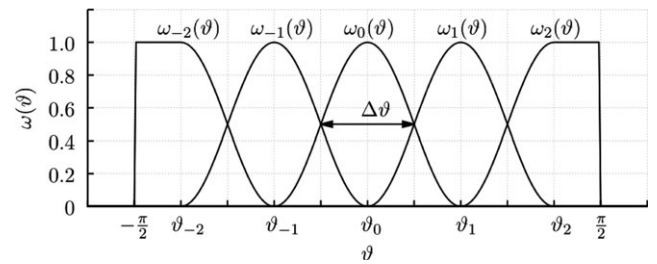


FIG. 1. Weighting function $\omega_k(\vartheta)$ applied to the segmented stacks projection data for PAR reconstruction illustrated for $K = 2$.

Fig. 2. The orange arrows point to deviations in PARs created from raw data to PARs created from synthetic data, which originate from inconsistencies introduced in the reconstruction of PARs using synthetic data by mixing projections corresponding to similar view angles, but different time points. This effect is mainly limited to the first and the last PARs, since for the reconstruction of the latter projections from similar angles, which however correspond to maximally differing time points are used. The severity of this dissimilarity strongly depends on the correspondence between motion direction and scan angle range. In the other sectors, the correlation between time and projection angle is mainly preserved. However, creating PARs from synthetic data is essential to the performance of the algorithm, since static objects or objects undertaking a different motion than the vessels, which are not contained in the ROI, still have an influence in projection space. Hence, in PARs directly obtained from raw data, those objects appear as streak bands (highlighted by the yellow arrows in Fig. 2). In the clinical case, the influence of such streak bands is non-negligible compared with the influence of temporal inconsistencies, which is why we reconstruct the PARs based on synthetic data. Furthermore, to omit inconsistencies, which can be caused by mixing of inconsistent projection views when using Parker weighting, we perform the initial reconstruction of $f_c(\Omega_s)$ based on rebinned data covering exactly 180° .

Since the PARs cover an angular range of $2\Delta\vartheta$ with a FWHM of $\Delta\vartheta$, we assume that the temporal window of each PAR is approximately reduced by a factor of $2K + 1$ compared with the original short-scan reconstruction.

We choose the number of PARs to be as large as reasonably possible. Since the motion model introduced in Section II.A.4 is independent of K , we choose to create PARs of high temporal resolution, which enables us to understand them as a quasi-static representation of the object of interest. For our PAMoCo reconstructions, we choose $K = 13$. Given the scanners rotation time of $t_{\text{rot}} = 285$ ms this yields a theoretical temporal window limit of 5 ms FWHM assuming perfect MoCo.

2.A.3. Motion Compensation

Since the PARs correspond to different time points, we obtain a motion-compensated reconstruction, determined by parameter set p , by adding the PARs, which have been

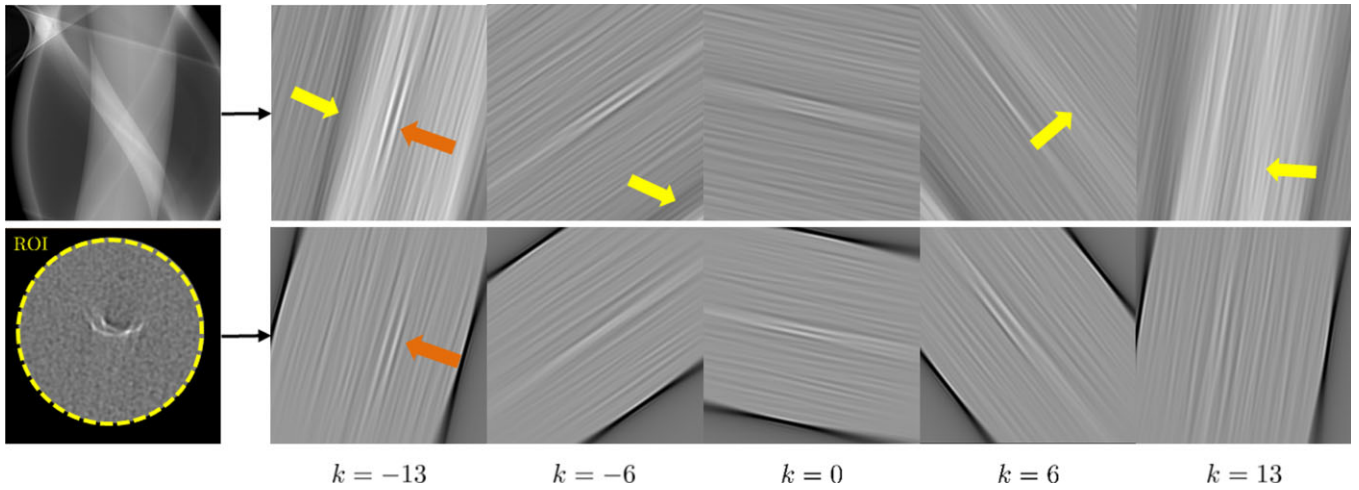


FIG. 2. The original, segmented FBP reconstruction of a phantom measurement and an excerpt of the raw data are shown in the left column. The contributions from nonmoving, or static background objects, in this case caused by the border of the spine of a body phantom, introduce bright streaks (light arrows) in PARs created from raw data (upper row), which are highly likely to influence the quality of PAMoCo. Creating PARs based on synthetic data derived from the segmented image allows removing those contributions in the PARs (lower row), while temporal resolution is still maintained. [Color figure can be viewed at wileyonlinelibrary.com]

modified with respect to the MVF $s(\mathbf{p}, \mathbf{r}, t)$:

$$f_{\text{MoCo}}(\mathbf{p}, \mathbf{r}) = \sum_{k=-K}^K f_k(\mathbf{r} + s(\mathbf{p}, \mathbf{r}, t)).$$

The new coordinate is given by a shift depending on voxel position $\mathbf{r}$ shifted by the motion vector $s(\mathbf{p}, \mathbf{r}, t)$. The parameters $\mathbf{p} \in \mathbb{R}^{N_p}$ model the motion and define the vectors s as described in Section 2.A.4. For a zero parameter vector $\mathbf{p} = \mathbf{0}$, f_{MoCo} corresponds to the original stack reconstruction $f_c(\Omega_s)$. Compared with FBP, the memory requirements are slightly increased when following this approach. However, it is possible to apply the MVF in a fast manner reducing computation time. After having determined the parameter vector $\mathbf{p}$, we insert the motion-compensated ROI f_{MoCo} back into the initial reconstruction f_c to obtain the final motion-compensated reconstruction for each stack.

2.A.4. Motion model

In order to keep the number of parameters low, one might think of modeling the temporal dependence of the MVF by a low degree polynomial as introduced by Hahn et al.³:

$$s(\mathbf{r}, t) = \sum_{i=1}^I \mathbf{a}_i(\mathbf{r})(t - t_0)^i = \omega(\mathbf{r}) \sum_{i=1}^I \mathbf{a}_i(t - t_0)^i, \quad (2)$$

where the vectors $\mathbf{a}_i(\mathbf{r})$ contain the motion modeling parameters. The spatial dependence in Eq. (2) is given by the weighting function $\omega(\mathbf{r})$ forcing a smooth transition of the MVF from the inner of the ROI to its borders. However, for large detector collimations long center line sections might pass through a stack and it might not be sufficient to shift the whole volume based on a single vector. As a consequence, it is reasonable to treat spatial and temporal dimension on equal terms. Hence, we introduce an additional polynomial,

incorporating a spatial evolution along the center line $\mathbf{r}_{\text{CL}}(\lambda)$ in λ . This is why we extend the parameterization introduced by Hahn et al.³ and model motion as well along the center line, which we parametrize by λ , with the help of a low-degree polynomial characterizing the temporal and spatial evolution of the vectors $s(\mathbf{p}, \lambda, t)$:

$$s(\mathbf{p}, \lambda, t) = \sum_{i=1}^I \sum_{j=0}^J \mathbf{p}_{ij}(t - t_0)^i (\lambda - \lambda_0)^j, \quad (3)$$

with $\mathbf{p}_{ij} \in \mathbb{R}^3$. This way the number of independent parameters $N_p = I \times (J + 1) \times 3$ is kept low and an indirect regularization of the MVF is implied.

Since we want to warp all PARs f_k with respect to the central phase at $t = t_0$, we demand $s(\mathbf{p}, t - t_0) \stackrel{!}{=} \mathbf{0}$. Since

$$s(\mathbf{p}, \lambda, t_0) = \sum_{j=0}^J \mathbf{p}_{0j}(\lambda - \lambda_0)^j,$$

we can start the sum over i in Eq. (3) at $i = 1$ to achieve that aim. With the general choice of $I \leq 4$, $J \leq 2$ we end up at $N_p \leq 36$ independent parameters, which we have to determine.

2.A.5. Motion vector field

In order to create a dense 3D MVF $s(\mathbf{p}, \mathbf{r}, t_k)$ for each PAR corresponding to time point t_k , we place N spatial control points within a distance of $|\mathbf{r}_{\text{CL}}(\lambda_{n+1}) - \mathbf{r}_{\text{CL}}(\lambda_n)| = 1$ cm along the center line. The 3D MVF

$$s(\mathbf{p}, \mathbf{r}, t) = \sum_n \omega_n(\mathbf{r}) s(\mathbf{p}, \lambda_n, t) \Big|_{d_n < R}$$

can be generated from the vectors derived along the center line by extrapolating $s(\mathbf{p}, \lambda, t)$, evaluated only at positions λ_n into the 3D volume. While doing so, we downweight the

influence of control points, which are further away than $2R$. The weighting function

$$\omega_n(\mathbf{r}) = T(d_n(\mathbf{r})) \frac{(d_n(\mathbf{r}) + \epsilon)^{-p}}{\sum_n (d_n(\mathbf{r}) + \epsilon)^{-p}},$$

calculates an effective vector applied at voxel position $\mathbf{r}$ using inverse distance weights with $d_n(\mathbf{r}) = d(\mathbf{r}, \lambda_n)$. In view of the computational performance, the distances, normalization, and the trapezoid weight can be calculated before optimization and used as lookup tables. Furthermore, for the analysis in this work we choose $P = 1$. To ensure that no voxel leaves the ROI, we furthermore introduce the trapezoidal function $T(d)$, with $T(d) = 1$ if $d < R$, $T(d) = 0$ if $d > 2R$ and linear in-between. Herein, we prevent singularities at the control point positions by adding $\epsilon = 0.3$ mm to the distances.

If we choose to place only a single control point $\lambda_1 \equiv \lambda_0$, which we place at $\mathbf{r}_{CL}(\lambda_0)$ — the center of a stacks center line — the new parameterization reduces to

$$\mathbf{s}(\mathbf{r}, t) = \sum_{n=1}^{N=1} \omega_n(\mathbf{r}) \mathbf{s}(\mathbf{p}, \lambda_0, t),$$

which is equal to the parameterization introduced by Hahn et al.³

2.A.6. Motion estimation

Since the algorithm is working with nothing more than the data needed for a single short-scan reconstruction and since no data redundancies for the reconstruction of further cardiac phases are present, it is not possible to determine the MVF defining parameters $\mathbf{p}$ employing an image registration routine. Instead, as proposed by Rohkohl et al.,¹⁷ an image artifact measuring cost function is subject to optimization. For such purpose we chose the entropy of the images:

$$\mathcal{C}(f(\mathbf{p})) = \sum_g P_{f(\mathbf{p})}(g) \ln(P_{f(\mathbf{p})}(g)).$$

The presented heuristic approach to optimization tries to balance the ability to determine a suitable MVF with a finite computation time. In our implementation the probabilities $P_{f(\mathbf{p})}(g)$ to find a gray value g are estimated by histograms calculated inside the part of the segmented region Ω_s , which will be replaced in the original FBP reconstruction. For $f(\mathbf{p})$ we will use two options:

1. $f(\mathbf{p}) = f_{\text{MoCo}}(\mathbf{p})$ being the motion-compensated image itself
2. $f(\mathbf{p}) = |\nabla f_{\text{MoCo}}(\mathbf{p})|$ being the absolute gradient motion-compensated image.

To diminish the issue of the cost function being nonconvex and in order to be able to also compensate for quite strong motion occurring during data acquisition (up to 1 cm as discussed in Section I), we divide the optimization routine into two parts allowing for a pseudo-global optimization

- (a). a brute force search combined with an optimization addressing only the three parameters $\mathbf{p}_{10}$, which will be used as initializing vectors for
- (b). the actual optimization, taking all parameters defined in Eq. (3) into account.

In part (a) we crudely assume the temporal dependence of the motion to be linear in time and neglect any dependence along λ , corresponding to $I = 1, J = 0$ in Eq. (3). Hence, we want to find the parameters $\mathbf{p}_{10} \in \mathbb{R}^3$ defining a constant vector field for each stack. For linear motion, the maximum expected shift equal to

$$s_{\max} = \left(R, R, \min\left(\frac{z_{\text{Lo}}}{2}, \frac{z_{\text{Hi}}}{2}\right) \right)$$

limits the parameter space $\mathbf{p}_{10} \in [-s_{\max}, s_{\max}]$, which we scan in steps of $2s_{\max}/N$. Note that the ability to compensate motion in the z -direction is limited by the overlap $z_{\text{Lo}, \text{Hi}}$ of the volumes. In order to limit the computational effort, we choose $N = 10$, since we need to apply the MVF and evaluate the cost function, for which we choose option 1), $N^3 = 1000$ times. Each time, we find a set of parameters leading to a lower entropy than the current best value, we start an optimization of the chosen cost function with those vectors and save the resulting parameters and entropy values for part (b), where we initialize an optimization using option 2) considering all parameters defined in Eq. (3) at the 10 best start vectors determined in step (a).

For fast optimization we used an implementation of Powell's algorithm,³¹ where we strictly constrain the parameter space to parameters leading to shifts smaller than $s_{\max}$. With the parameterization defined in Eq. (3) it is important to choose meaningful initial step lengths considering the scale on which the parameters operate. The initial step lengths we chose decrease with the increasing total power $i + j$ of the polynomial presented in Eq. (3). Furthermore, to prevent too long computation times, we stop the iteration process, if the cost function values do not improve more than 10^{-4} and allow a maximum of 100 iterations in each optimization.

2.B. Evaluation

2.B.1. Phantom measurement

For our analysis we performed a phantom measurement with a Siemens SOMATOM Force DSCT system with $t_{\text{rot}} = 250$ ms in a low-pitch spiral acquisition mode, which enabled us to reconstruct multiple cardiac phases and evaluate the performance of the algorithm facing motion of different amplitude and severity. For motion compensation we used only the data needed for the reconstruction of a single cardiac phase. Furthermore, since our work focuses on single-source systems, we used only the data acquired with one detector system for reconstruction.

To create images with a quality comparable to clinical scans, we chose a tube current of 250 mA and a peak voltage of 100 kV. The actual vessel phantom is a rigid, artificial

artery made of hydroxyapatite shown in Fig. 3 (made by QRM, Möhrendorf, Germany). It consists of a synthetic cylinder of 4 mm diameter imbedding three calcium lesions covering 50%, 25%, and 12.5% of the lumen of the affected vessel sections. The hydroxyapatite concentration of all lesions is 796 mgHA/cm³ corresponding to high-density plaques with a density of 1.58 g/cm³. With the chosen acquisition settings this density corresponds to about 1100 HU in a static reference CT image reconstruction. The free vessel lumen corresponds to a contrast-enhanced vessel with CT values around 300 HU. To create realistic scanning conditions, the artificial artery was placed inside a water tank surrounded by a thorax phantom simulating a slim patient, as shown on the right side of Fig. 3. A heart rate-dependent motion was applied by attaching the artery to a motion robot, which also supplied the CT system with an ECG signal needed for cardiac imaging. The simulated motion at 70 bpm, shown in Fig. 4, and 90 bpm takes the nonproportional shortening of diastole, whose plateau phase starts vanishing at higher heart rates, into account as also described by Hahn et al.⁴ Based on RCA velocity curves known from literature,^{1,5,22,29} a reduction of the motion amplitude at higher heart rates has been taken into account in addition to the shortening of the diastolic phase, as shown in Fig. 5. For the 70 bpm and 90 bpm

measurements the pitch was adapted to the heart rate to 0.23 and 0.14, respectively.

In order to measure the performance of the algorithm, we measured the diameter d of the vessel and computed the volume V of the calcification before and after applying PAMoCo and compared both results to the ground truth (GT) diameter and volume. Measures such as the volume are also used for the computation of calcium scores,³⁰ which are highly susceptible to motion and hence we also expect to see strong variations over the heart cycle in case of the FBP especially for the high-density calcification. If we are able to eliminate motion artifacts in the reconstructions, we should be able to restore the object's shape and expect to determine diameters and volumes close to the true values for many heart phases. Besides analyzing the shape of the object of interest after PAMoCo, we also computed the center of mass positions (x,y) in the FBP and PAMoCo reconstructions, to validate the vessel position after applying PAMoCo.

For the computation of V we defined an ROI around the largest lesion covering 50% of the vessel lumen with a field of view (FOV) covering 20 mm, which is shown in Fig. 3. For any reconstruction, in which we wanted to evaluate V , we moved the ROI, such that the largest inclusion was covered and excluded the others lesions from our examination. For

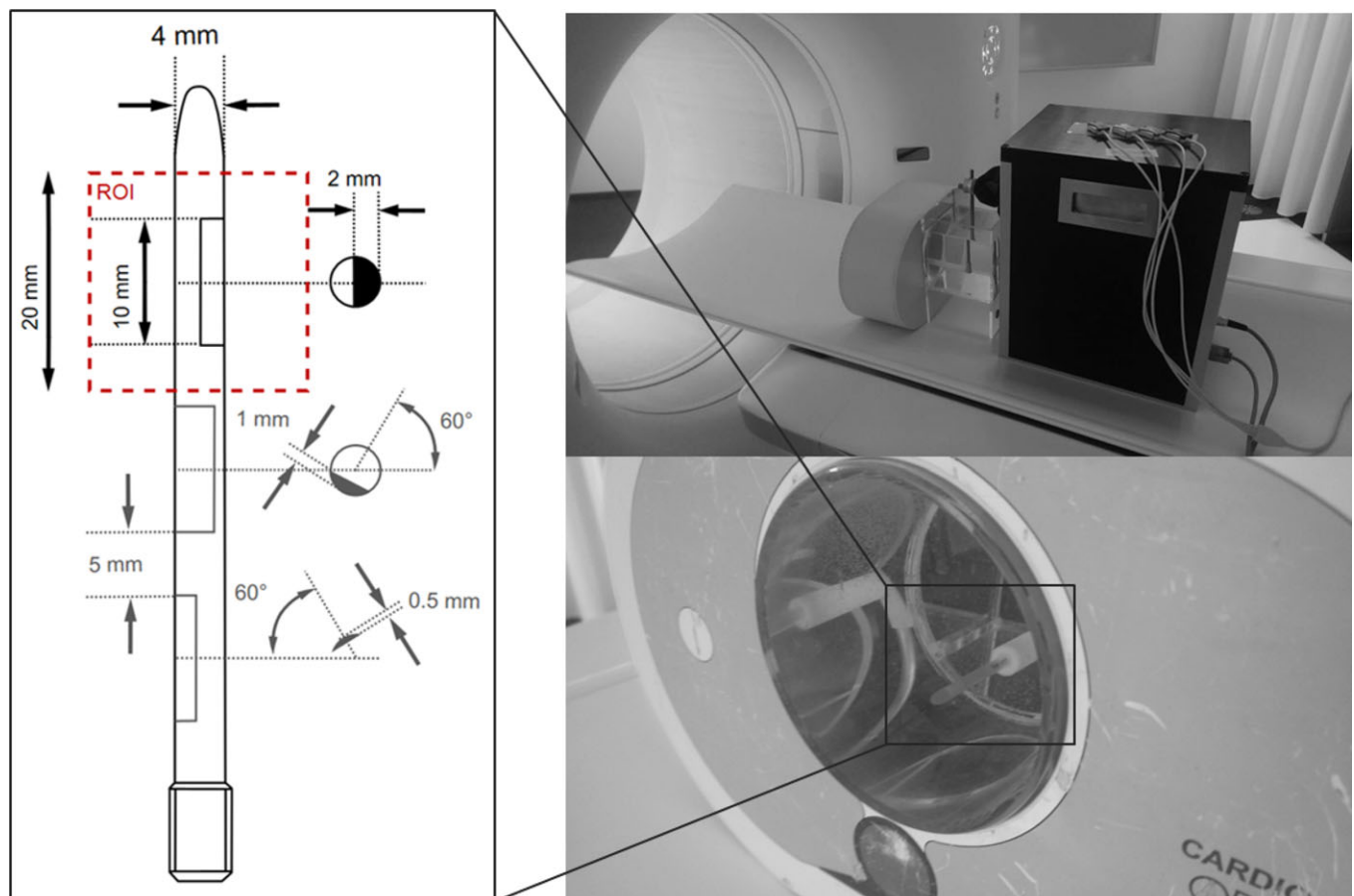


FIG. 3. We calculate the calcified volume V and measure the maximum vessel diameter d inside a cubic ROI with a width of 20 mm covering the largest calcification of the synthetic vessel. [Color figure can be viewed at wileyonlinelibrary.com]

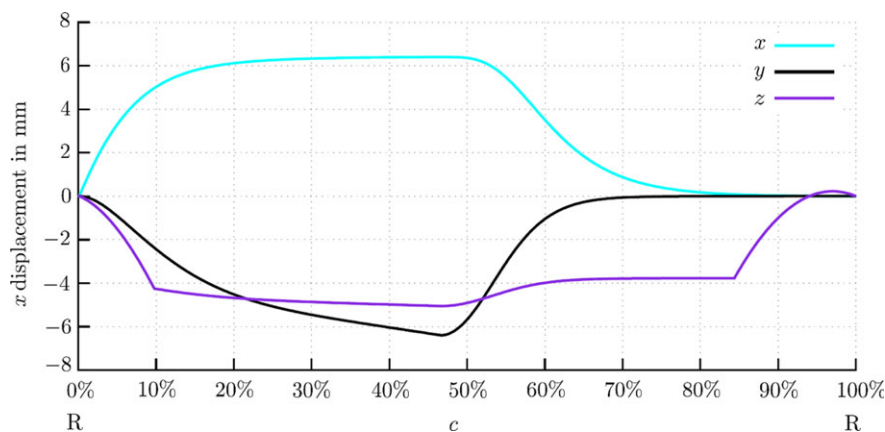


FIG. 4. x, y, z - displacement of the vessel phantom over a cardiac cycle for a simulated heart rate of 70 bpm. [Color figure can be viewed at wileyonlinelibrary.com]

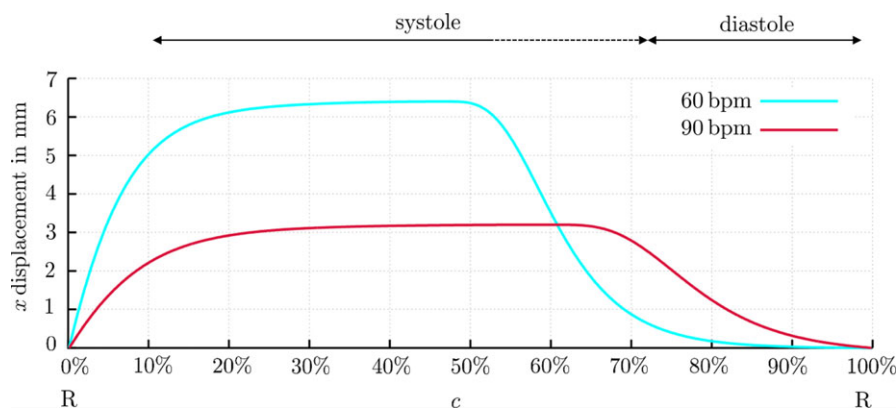


FIG. 5. Motion phantom displacement during an RR-interval in mm in x -direction at 70 and 90 bpm. With increasing heart rate, a lowering of the amplitude and shortening of the diastolic phase, as indicated by the dashed line, has been taken into account. [Color figure can be viewed at wileyonlinelibrary.com]

each ROI, we identified the volume V of the largest lesion, for which we performed a simple threshold-based segmentation, leaving connected regions with CT values larger than a threshold $CT_{thres} = 420$ HU excluding water and contrast-enhanced vessel from the calculation. Furthermore, we measured the diameter and calculated the center of masses from the center axial planes of the ROI of the FBP and PAMoCo images before applying the threshold, where we define the diameter as the maximum section length through the gray value distribution, associated with the object of interest.

2.B.2. Clinical cases

We further evaluated the algorithms with six coronary CTA patient cases with heart rates between 50 bpm and 74 bpm acquired with Siemens SOMATOM Definition systems, with a rotation time of $t_{rot} = 285$ ms in a low-pitch spiral mode. The temporal resolution of all FBP reconstructions can be approximated by $t_{res} \approx 143$ ms. The acquisition settings, such as tube voltage values and tube current of the respective patients, and pitch values leading to varying image qualities of the initial FBP reconstructions, are listed in Table I. Furthermore, to obtain difficult initial FBP

TABLE I. Parameters of clinical cases used for the performance evaluation of the proposed MoCo algorithm.

Patient	Mean heart rate	Stented	CT system	Tube current	Tube voltage	Pitch
1	74 bpm	Yes	Flash	656 mA	120 kV	0.23
2	70 bpm	No	Edge	657 mA	120 kV	0.23
3	50 bpm	No	Edge	464 mA	80 kV	0.21
4	59 bpm	Yes	Edge	399 mA	100 kV	0.17
5	59 bpm	No	Edge	531 mA	80 kV	0.21
6	66 bpm	No	Edge	445 mA	80 kV	0.20

reconstructions featuring strong motion artifacts, we chose to reconstruct all cases at $c = 50\%$ of the cardiac cycle.

3. RESULTS

3.A. Phantom measurement

We performed standard single-source FBP and PAMoCo reconstructions at heart phases between 20% and 80% of the

cardiac cycle in increments of 2% of the cardiac cycle for motion simulated at 70 bpm and 90 bpm. In total, we performed 122 FBP and 122 PAMoCo reconstructions. Some axial slices from a subset of the reconstructed FBP and PAMoCo volumes for the measurement at 70 bpm and 90 bpm are displayed in Figs. 6 and 7, showing clipped images of the respective reconstructions.

From visual inspection, we can already see a reduction of motion artifacts when applying PAMoCo compared with the original FBP reconstruction. This applies without limitation to cardiac phases leading to not too severe motion artifacts. Also in cases when strong motion artifacts are present, for example, in the phases $c = 40\%$ – 60% in Fig. 6 and $c = 20\%$ – 28% and $c = 64\%$ – 80% in Fig. 7 a strong reduction of the latter can be observed. Furthermore, the image quality of already ‘good’ reconstructions (see phases $c = 20\%$ – 32% in Fig. 6 and $c = 32\%$ – 60% in Fig. 7) is not deteriorated after MoCo.

Those findings are also supported by the diameter and volume evaluations depicted in Figs. 8 and 9, which have been derived as described in Section II.B.1. The ground truth volume $V_{GT} = 66 \text{ mm}^3$ used in Fig. 9 used as a reference was derived based on a DSCT reconstruction visually featuring no motion artifacts, using the same method for the calculation of the volume. Both distributions, the diameter as well as the volume, agree well with the impression gained by visual inspection of the images of the occurrence of motion artifacts shown in Figs. 6 and 7.

Assuming no motion is present, we expect constant values throughout all cardiac phases. Motion artifacts, which

are visually tearing the vessel apart by distributing the gray values of an object over a larger region, will reduce V and increase d . Concerning the volume evaluated on the basis of the FBP reconstructions displayed in Figs. 6 and 7, we observe strong deviations from a constant value in cardiac phases, which we have identified as comprising motion artifacts in case of both heart rates. When using PAMoCo for the reconstruction, we observe V and d to be almost constant. In case of cardiac phases featuring little motion artifacts, V and d agree well with the GT values. In case of strong motion, slight deviations from the GT can be observed. However, a significantly better computation was possible using PAMoCo results compared with the original FBP.

Understanding the plots of V and d for FBP and PAMoCo reconstructions as probability distributions, we can calculate a mean value and a standard deviation σ . The reduction in σ by 82% (from 9.1 mm^3 to 1.6 mm^3) at 70 bpm and by 62% (from 10 mm^3 to 3.8 mm^3) at 90 bpm reflects the improvement in image quality consistency over all cardiac phases obtained by PAMoCo. This is also reflected by the diameter measurements leading to 95% and 92% reduction in σ for the 70 bpm and respectively the 90 bpm measurement. In summary, we were able to calculate consistent volumes and diameters, when using PAMoCo, which is caused by the considerable reduction of motion artifacts leading to a better discrimination from calcium to vessel and water background. The dependence of V and d on the reconstruction phase diminished, meaning that PAMoCo performed almost equally well for all investigated phases.

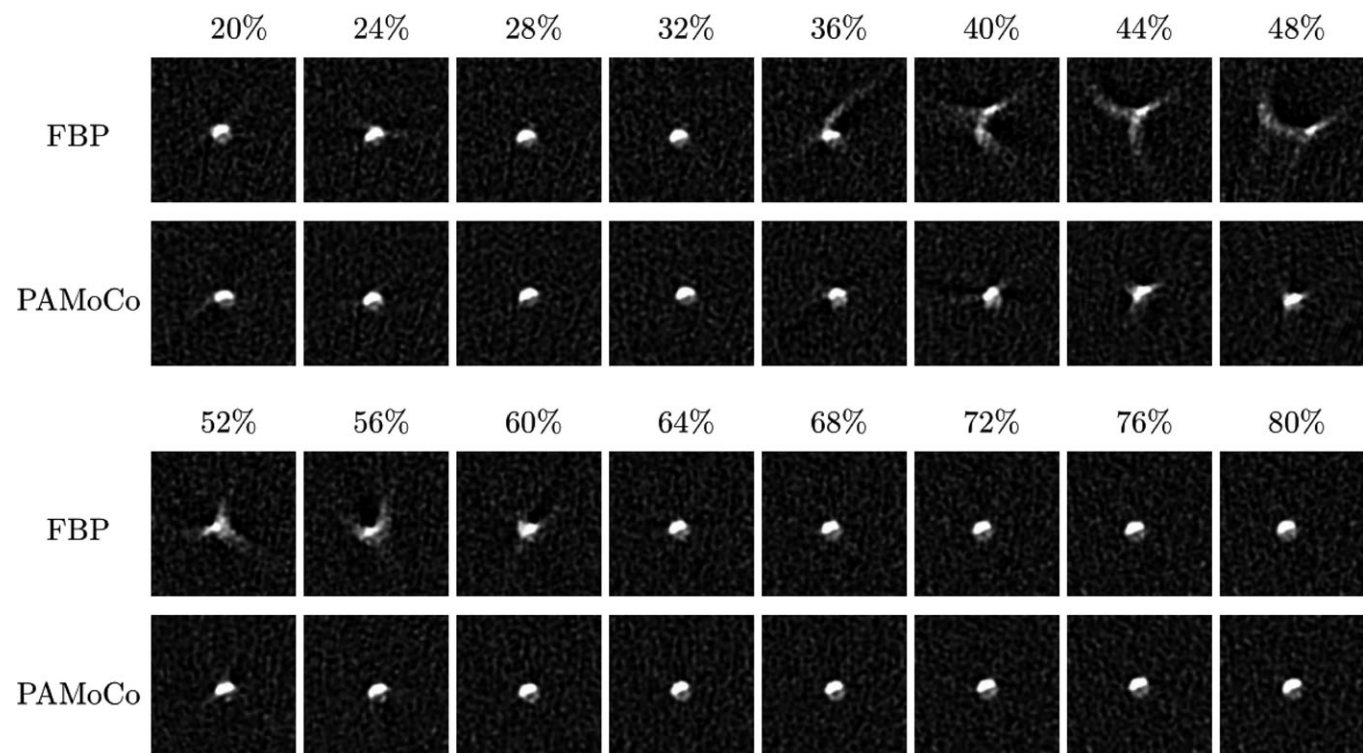


FIG. 6. FBP and PAMoCo reconstructions of the measured phantom featuring calcifications at an applied heart rate of 70 bpm at reconstruction phases c between 20% and 80%.

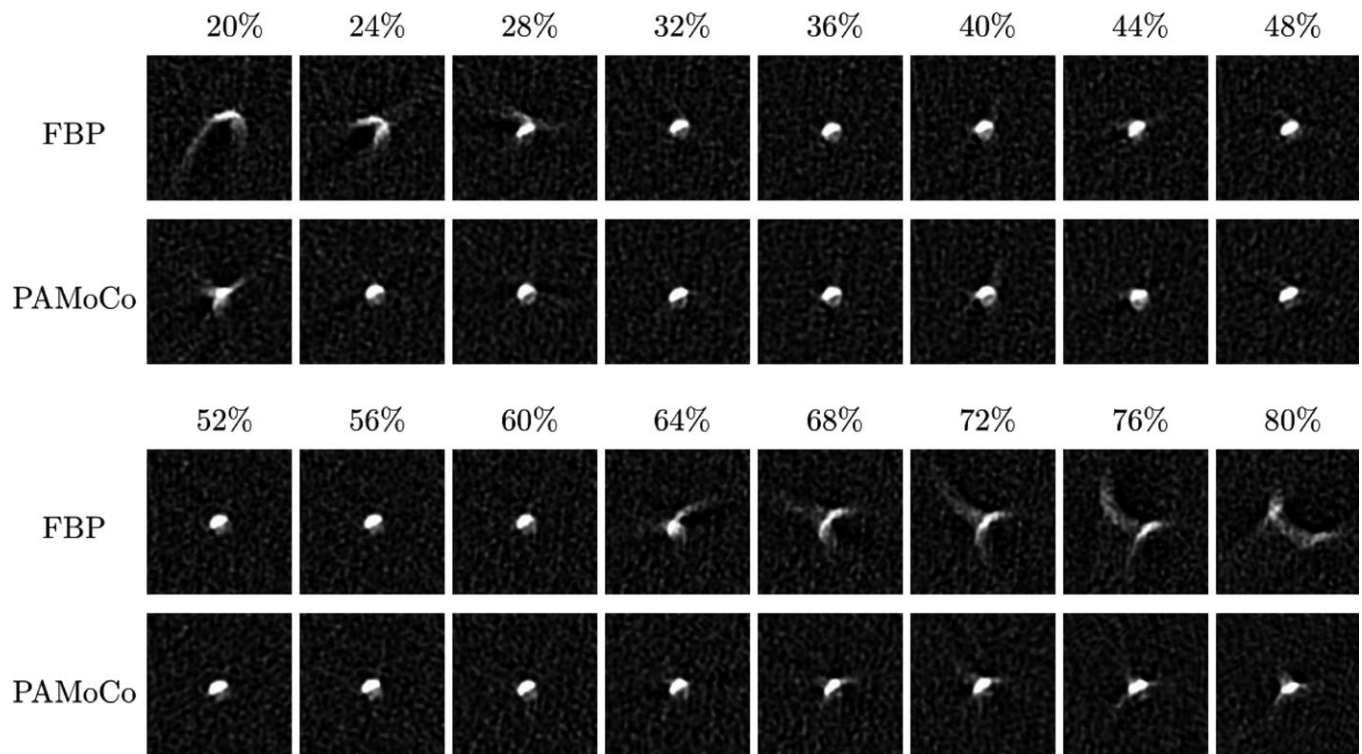


FIG. 7. FBP and PAMoCo reconstructions of the measured phantom featuring calcifications at an applied heart rate of 90 bpm at reconstruction phases c between 20% and 80%.

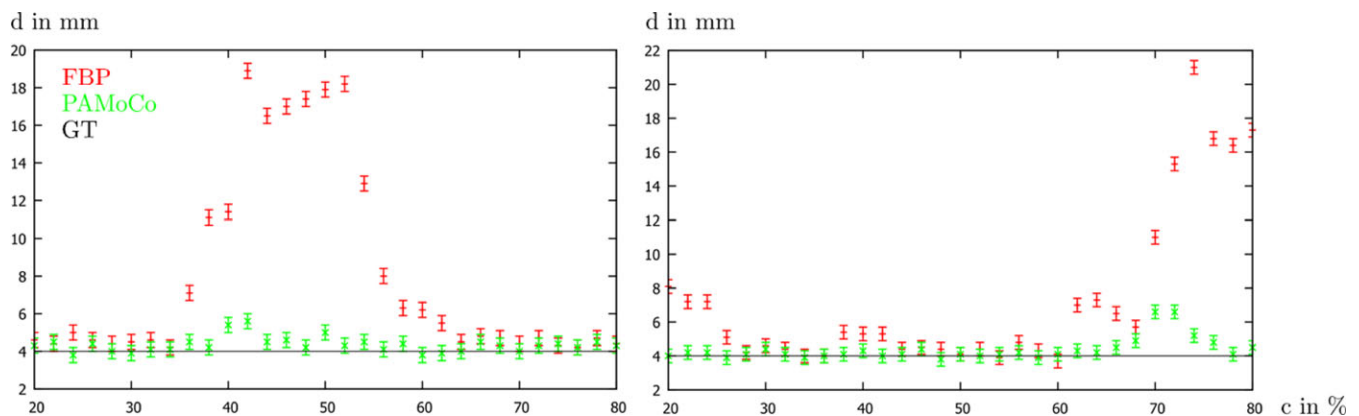


FIG. 8. In case of motion phases featuring strong in the FBP images, the maximum vessel diameter deviates strongly from the ground truth value (GT) of 4 mm in case of the simulated heart rates of 70 bpm (left) and 90 bpm (right). The error bars is Fig. 8 result from reading errors estimated to approximately 0.8 mm due to the manual diameter metering. [Color figure can be viewed at wileyonlinelibrary.com]

Finally, in Figs. 10 and 11 we provide the center of mass measurements for FBP and PAMoCo reconstructions over the selected cardiac range for the two measurement series. In addition to the positions x and y we show the center of mass differences $\Delta x = x^{\text{PAMoCo}} - x^{\text{FBP}}$ in FBP and PAMoCo reconstructions. The positions only significantly deviate from each other in cardiac phases featuring strong motion artifacts, but follow a smoother path, including less outliers in case of the PAMoCo reconstructions. The mean relative deviations $\bar{\Delta x}$ and $\bar{\Delta y}$ in both cases lie between 0.8 and 1 mm in the range of only a few pixels and are much smaller than the diameter of the measured object, which is why we consider the center of masses to be consistent.

The findings of the phantom measurements imply that the method is rather independent of the choice of reconstruction phase. Hence, in our further analysis, we paid special attention to cardiac phases featuring strong motion artifacts, where we expect to see most notable improvements in coronary artery delineation.

3.B. Clinical cases

In contrast to the phantom measurement, the clinical data give us the opportunity to test the algorithm for a more complex task. Besides the fact that in a real patient, the arteries are not surrounded by water, but different tissues, contrast-

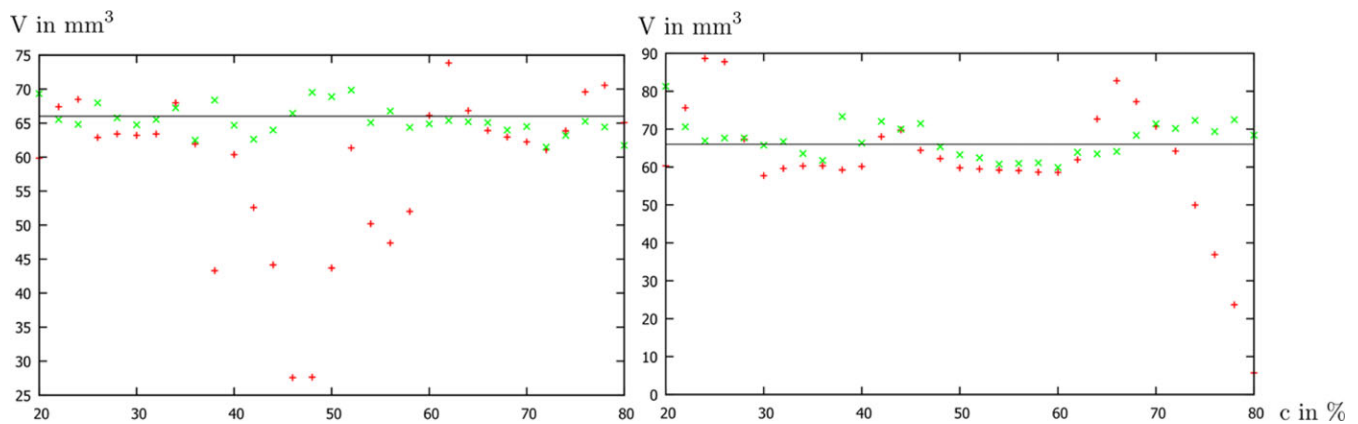


FIG. 9. The calculated volumes are also a suitable image quality measure, since they correlate well with the visual impression of motion artifacts occurring in the reconstructions. In both simulations (70 bpm shown on the left and 90 bpm shown on the right) it is possible to derive the volume more precisely using PAMoCo images. [Color figure can be viewed at wileyonlinelibrary.com]

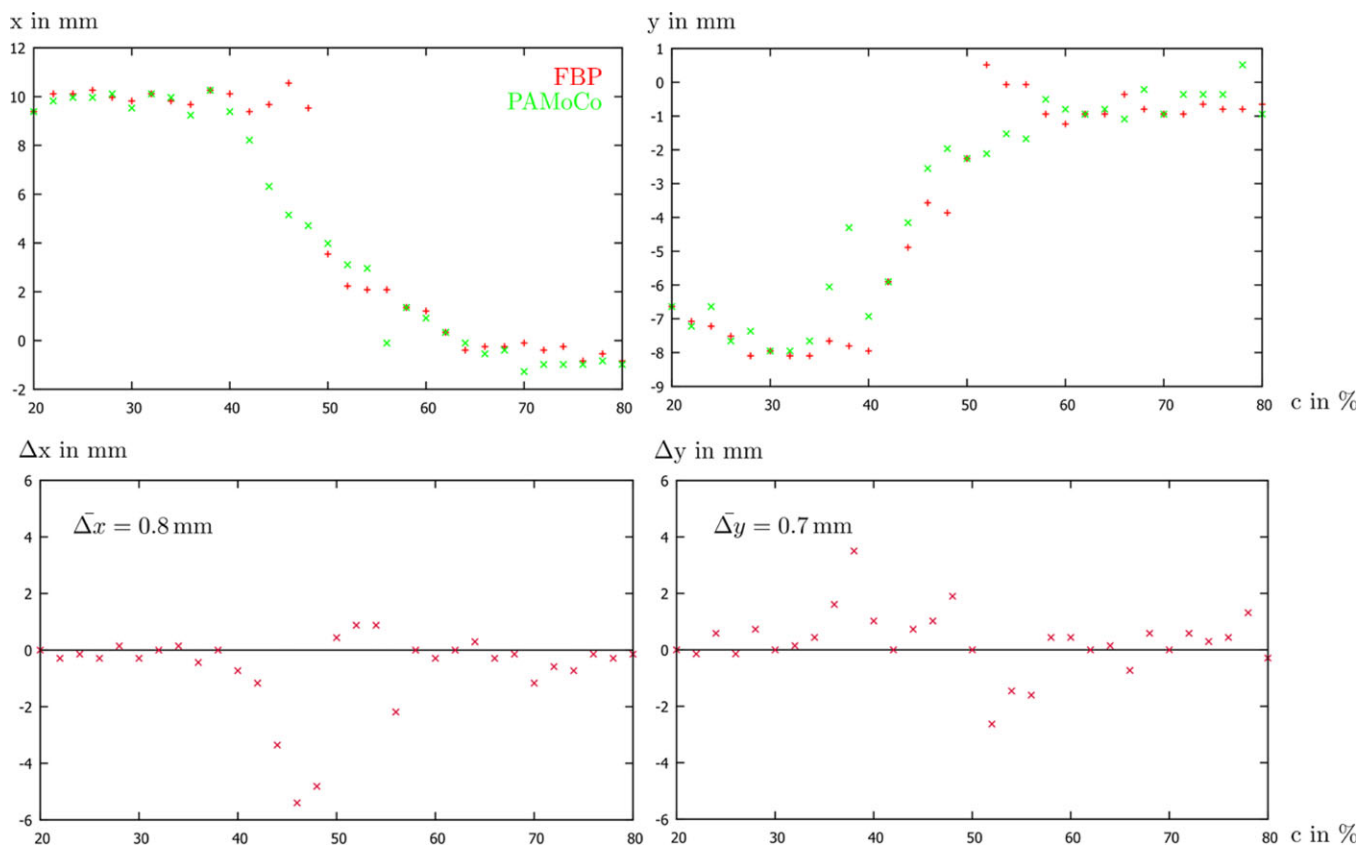


FIG. 10. Center of mass x and y positions in mm for FBP and PAMoCo images and relative deviation of the latter for a simulated heart rate of 70 bpm. [Color figure can be viewed at wileyonlinelibrary.com]

enhanced lipids, air, or even parts of bone. Further, anatomy differs for every patient, for example, the size of the coronary arteries varies.

For each patient, we performed an FBP and PAMoCo reconstruction improving the temporal resolution of the RCA at $c = 50\%$ of the cardiac cycle. Furthermore, we performed reconstruction of the same phase using the MAM algorithm mentioned in Section I and a best-phase FBP reconstruction, which we manually selected. For comparison of the

reconstructions, we chose 6 patient cases, all with an initial temporal resolution of 142.5 ms, with scan settings displayed in Table I, leading to varying initial image qualities. We chose to visualize the volumes in different views, such as axial slices (Figs. 12–17) and double-oblique views (Figs. 12 and 13), where motion artifacts along the right coronary artery are well visible. Furthermore, we created curved multiplanar reformatting (MPRs) (Figs. 18–20) and volume renderings of the coronary artery tree (Figs. 21–23) associated with patients

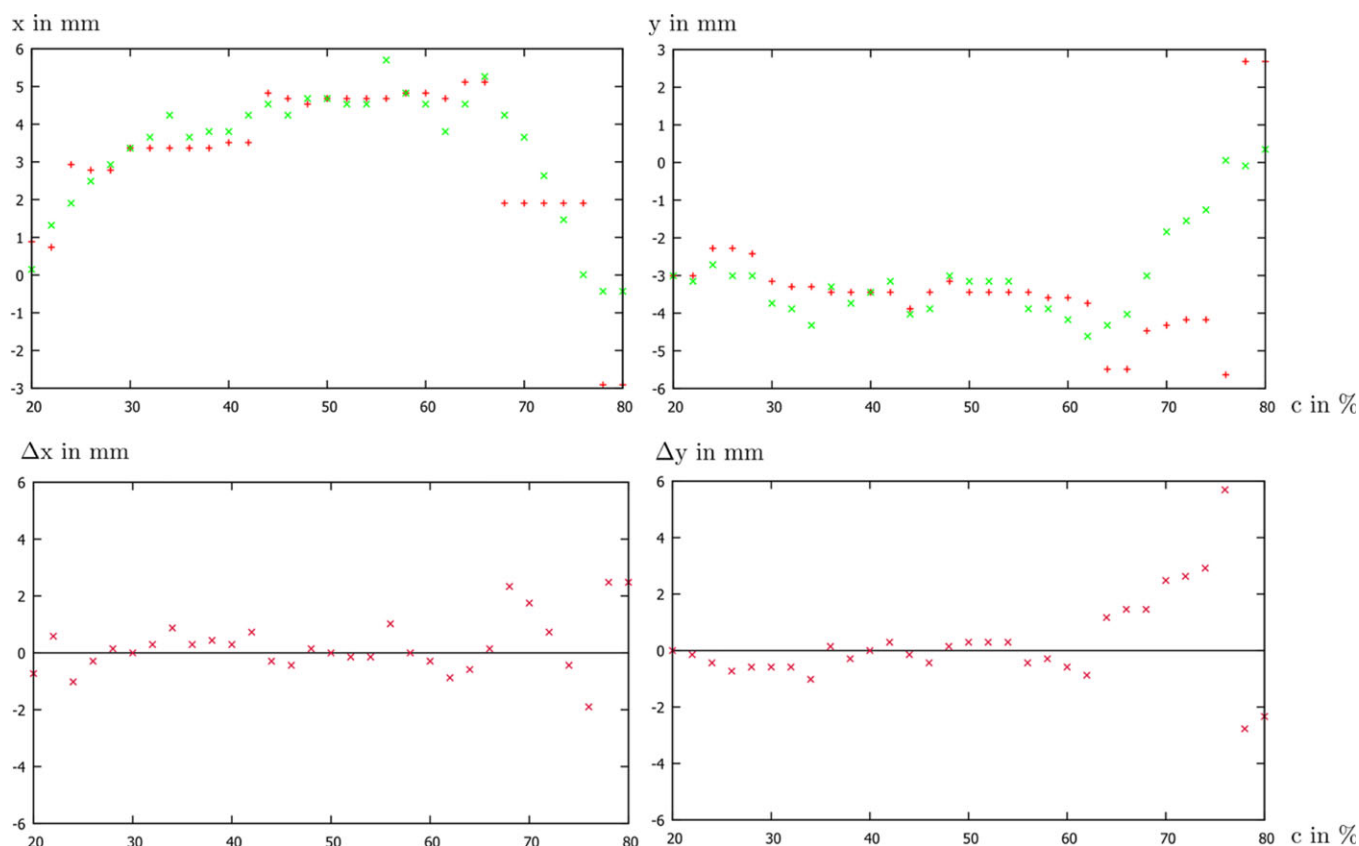


FIG. 11. Center of mass x and y positions in mm for FBP and PAMoCo images and relative deviation of the latter for a simulated heart rate of 90 bpm. [Color figure can be viewed at wileyonlinelibrary.com]

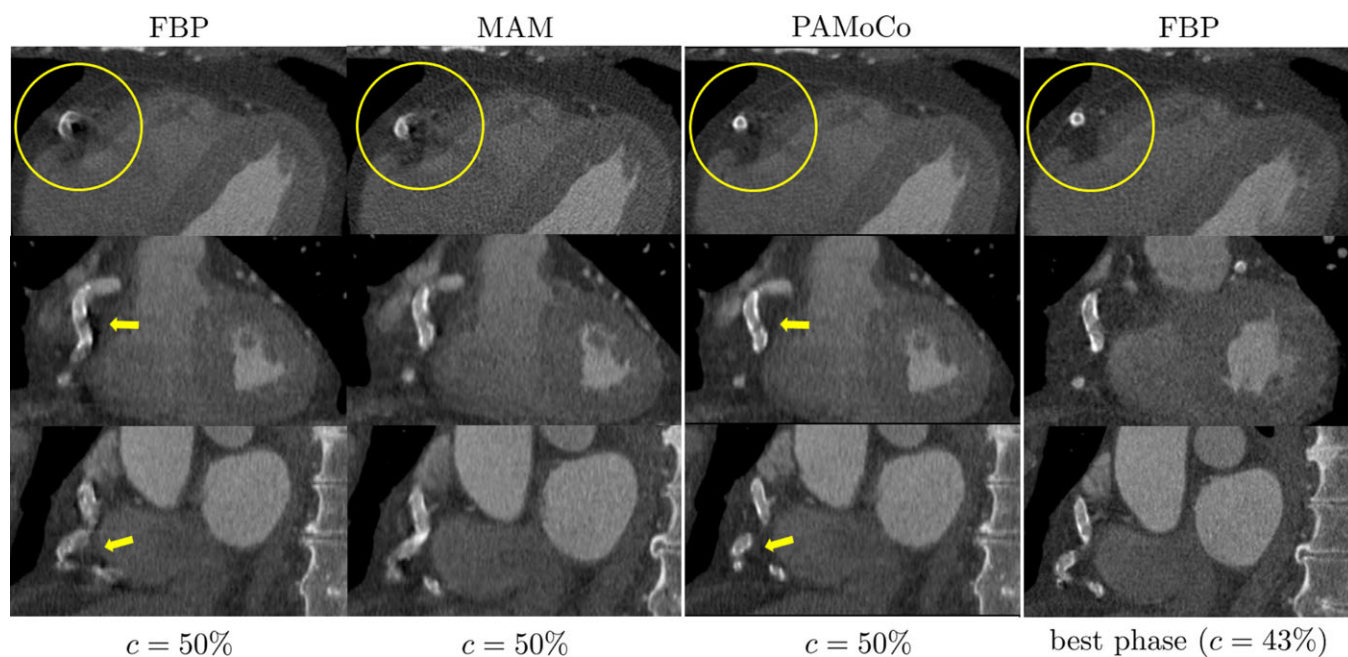


FIG. 12. Patient 1: FBP (first column) and PAMoCo reconstructions (third column) of a large, dilated RCA equipped with several stents displayed in an axial and two double-oblique views (top to bottom row) showing a reduction of motion induced shading artifacts when applying PAMoCo. The result is close to the best phase reconstruction (fourth column) and much better than the MAM result (second column). [Color figure can be viewed at wileyonlinelibrary.com]

1, 2, and 3. The volume renderings include the segmented coronary artery center line, which were used for the creation of the curved MPRs. Those were created with the

Syngo®.via (SIEMENS Healthcare GmbH) software, with which we were able to perform a segmentation on FBP and PAMoCo reconstruction separately. In cases, where motion

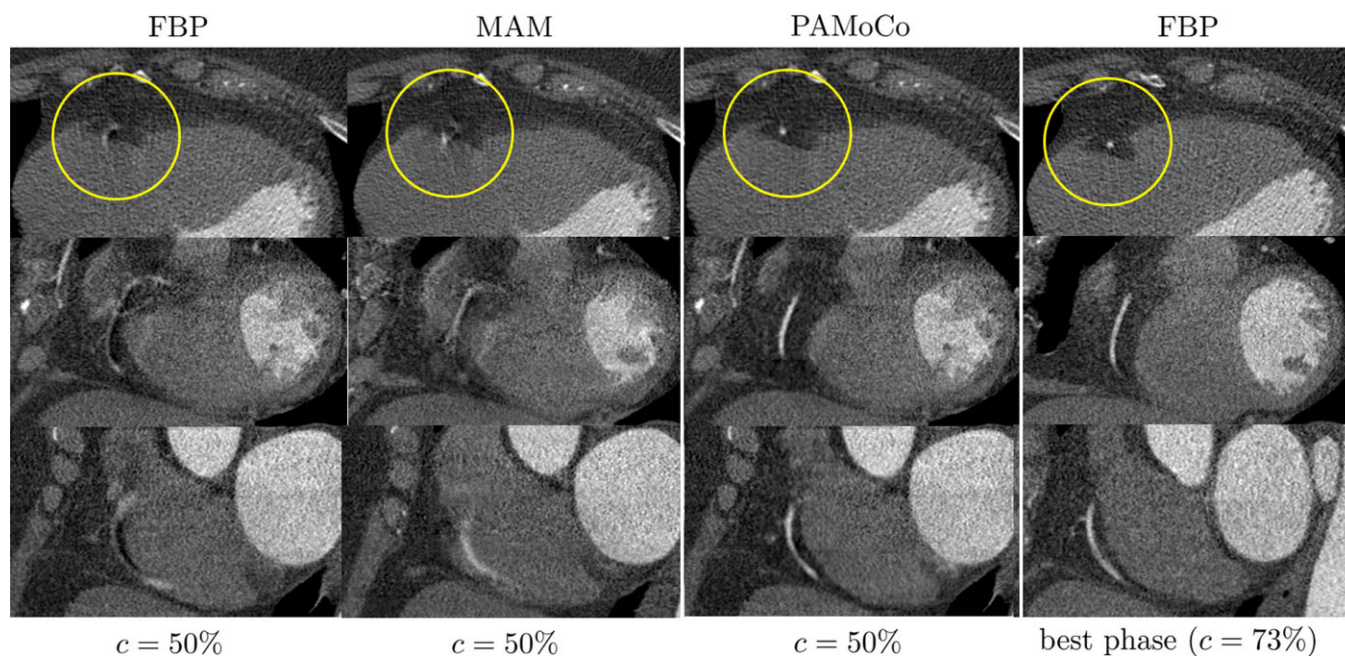


FIG. 13. Patient 2: FBP (first column) and PAMoCo reconstructions (third column) of a RCA of small diameter displayed in an axial and two double-oblique views (top to bottom row) showing a strong reduction of motion artifacts when applying PAMoCo. [Color figure can be viewed at wileyonlinelibrary.com]

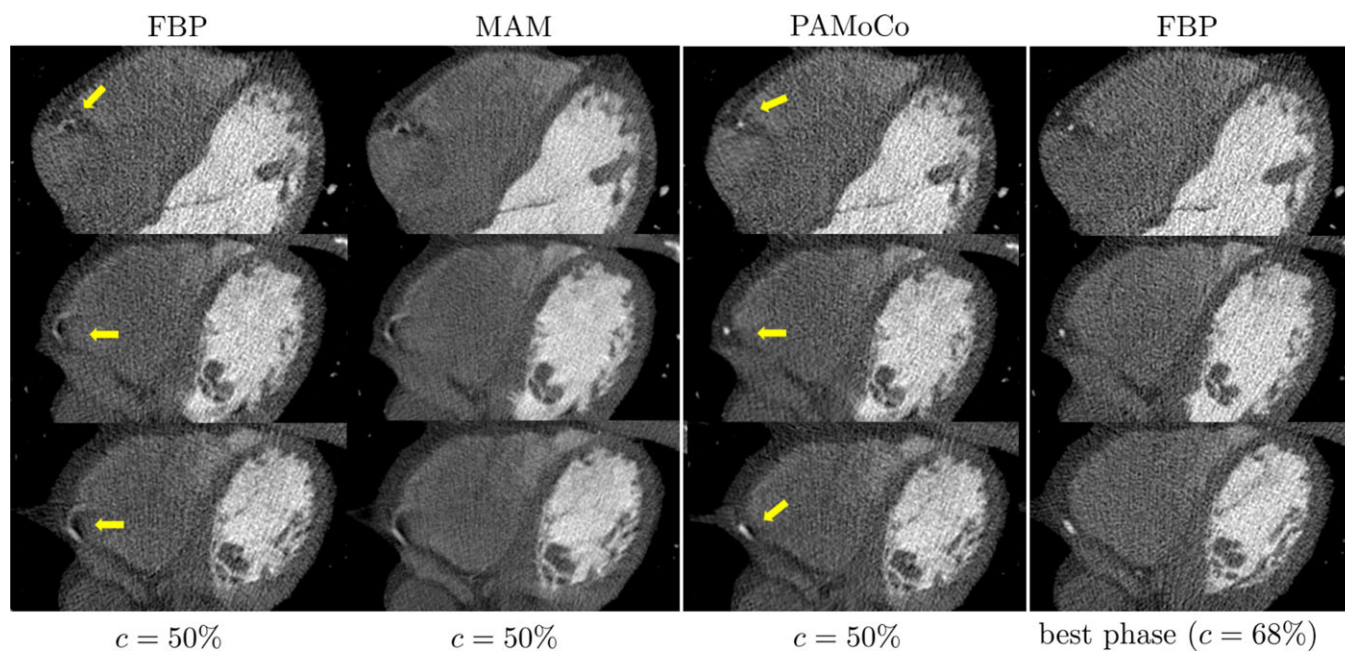


FIG. 14. Patient 3: FBP (left column) and PAMoCo reconstructions (right column) of a RCA of small diameter displayed in three axial views (top to bottom row) of a patient with slight arrhythmia, showing a strong reduction of motion artifacts when applying PAMoCo. [Color figure can be viewed at wileyonlinelibrary.com]

artifacts were too strong, such as in Fig. 21, the software was not able to detect the center line and hence partial segments are missing in the VR image. However, we manipulated the center lines by adding control points to create curved MPRs of FBP and PAMoCo reconstructions of similar view, for example, shown in Fig. 18. In one case the MAM result was so of such bad image quality, that we did not create corresponding VR and curved MPR images.

When applying PAMoCo we were able to diminish typical motion artifacts, for example, the doubling of the RCA, which is well visible in the volume rendering of the coronary tree and the curved MPR of the RCA based on the FBP reconstruction of patient 2 shown in Figs. 22 and 19. In general the vessels can be better discriminated from the background and appear much smoother, which is apparent in the volume renderings shown in Figs. 21–23. The improved

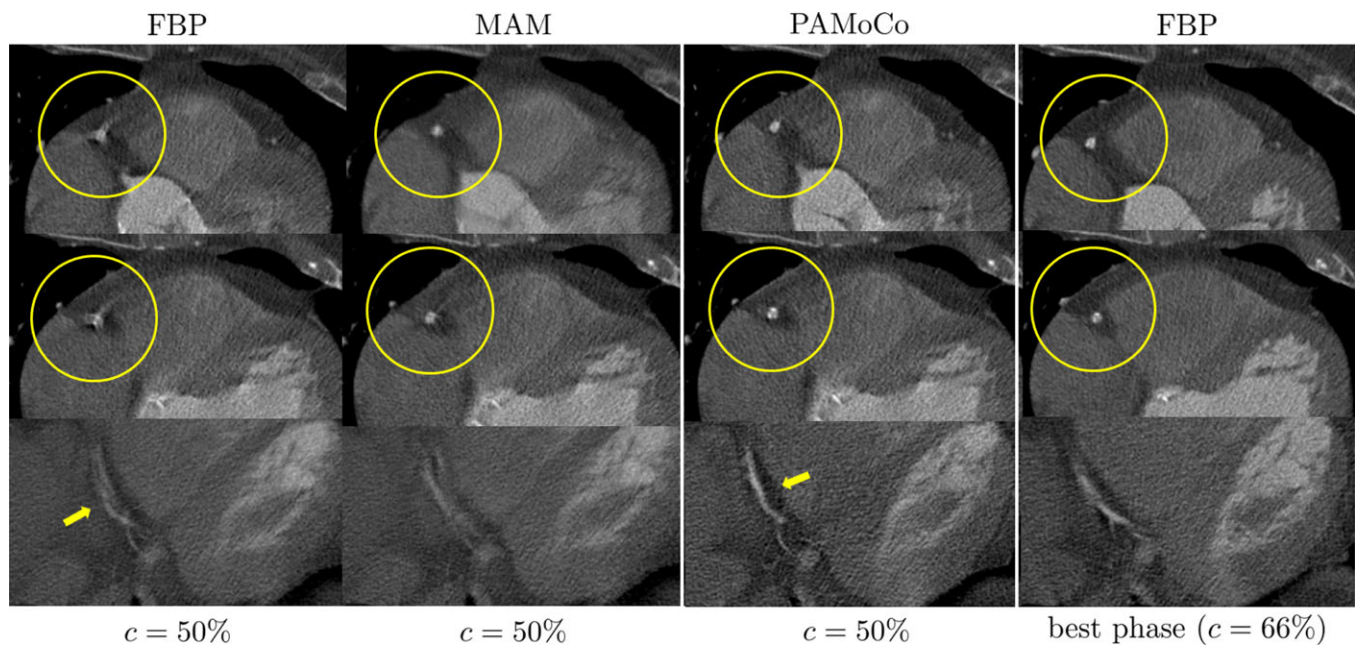


FIG. 15. Patient 4: FBP (first column) and PAMoCo reconstructions (third column) of a stented RCA of medium diameter displayed in axial views. Motion-induced blurring artifacts are diminished and the stent mesh becomes visible in the PAMoCo case. [Color figure can be viewed at wileyonlinelibrary.com]

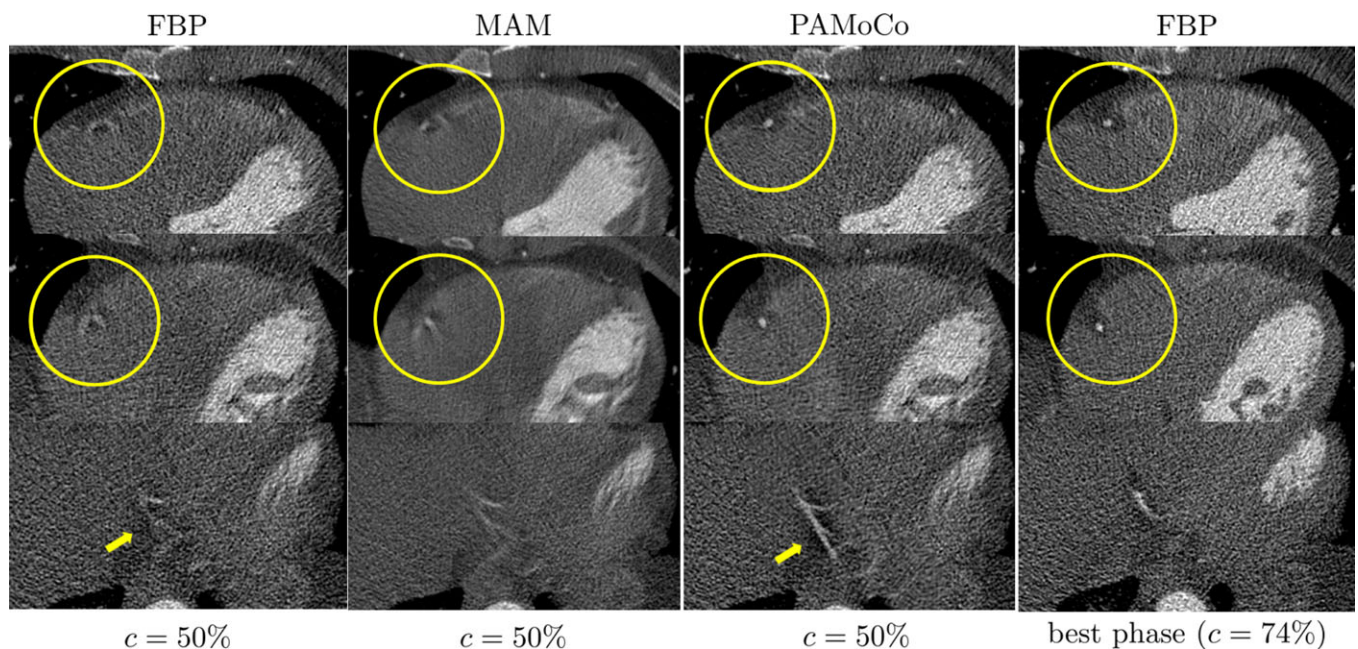


FIG. 16. Patient 5: FBP (first column) and PAMoCo reconstructions (third column) of a RCA displayed in axial views. Despite the high image noise and strong motion artifacts, the algorithm was able to increase the delineation of the artery. [Color figure can be viewed at wileyonlinelibrary.com]

visualization of the vessel lumen could also be beneficial for FFR CT measurements. Independent of the quality of the initial reconstruction, we were able to diminish strong motion artifacts, which we see when we compare for example FBP and PAMoCo reconstructions of patient 1 shown in Fig. 12, which has the highest image quality of the selected cases and patient 3 shown in Fig. 14 with one of the highest noise level. In both cases, motion artifacts were reduced considerably. Furthermore, note that in Fig. 20, due to slight irregularities

of the heart beat, the borders of the stacks mentioned in Section I are clearly visible. In Figs. 15–17 we observe that the size of the respective RCAs decreases. Image quality has improved in all of these cases and even very small features of little branches close to the segmented artery also benefit from MoCo since also small bifurcations reappear in front of the background. In Figs. 13 and 15 we see that stent structures become clearly visible in the PAMoCo reconstructions. Note further, that this is also the case for varying size of the RCA.

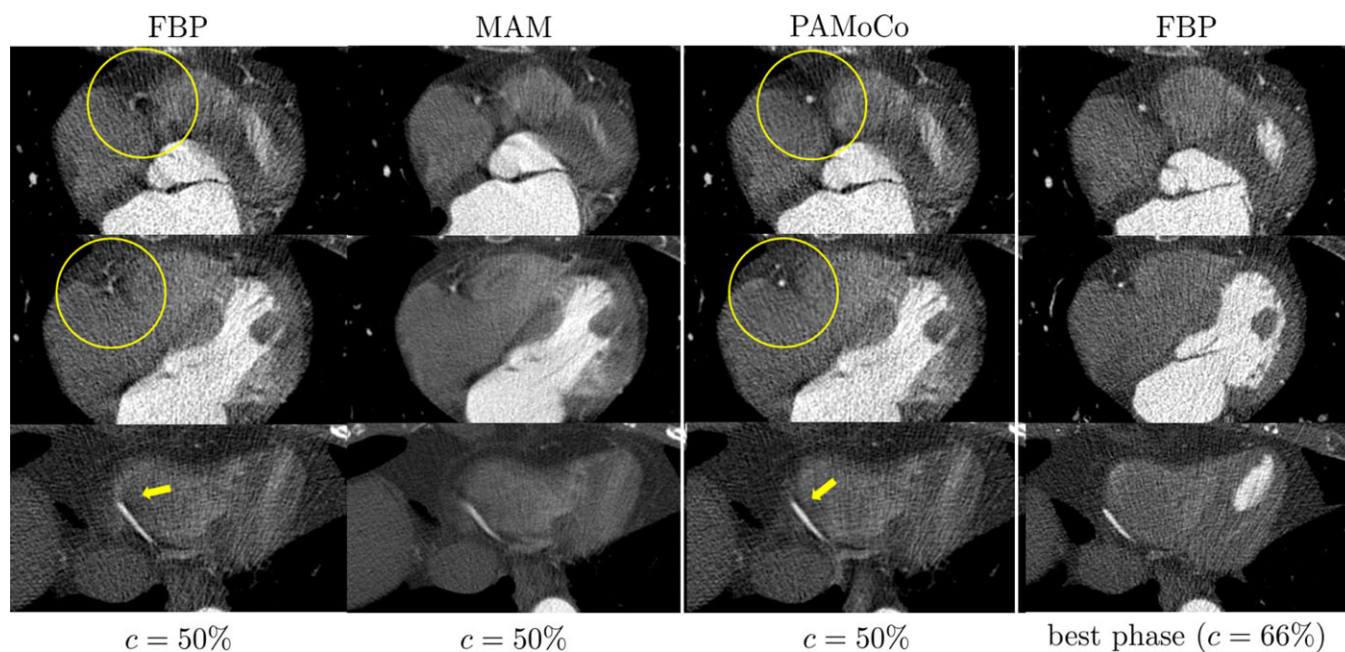


FIG. 17. Patient 6: FBP (first column) and PAMoCo reconstructions (third column) of a small RCA displayed in axial views. Even small features could be restored in the PAMoCo case. [Color figure can be viewed at wileyonlinelibrary.com]

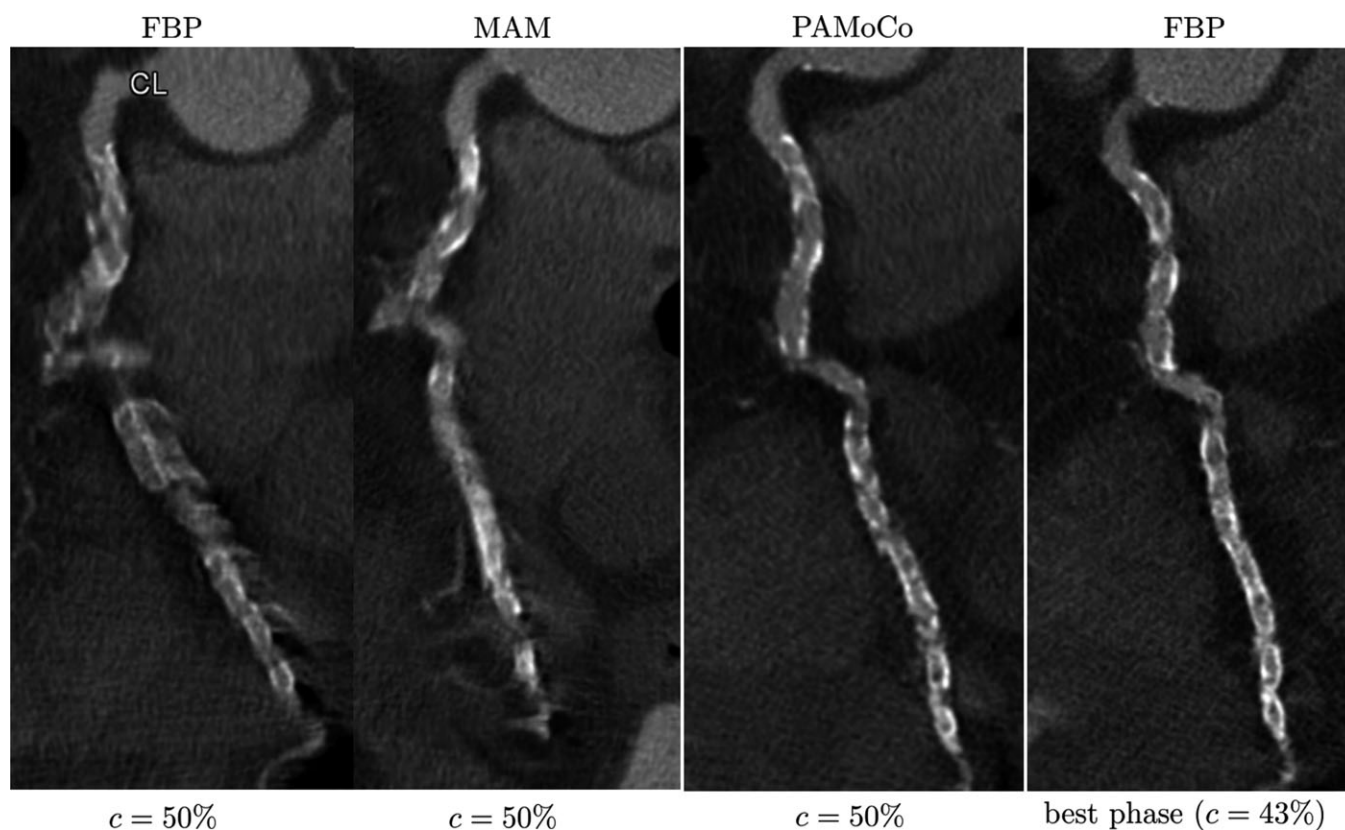


FIG. 18. Patient 1: The curved MPRs show a strong reduction of motion artifacts in the PAMoCo reconstruction (third from left) when compared with the original FBP reconstruction (first from left). However, the best phase image appears slightly sharper.

Overall image quality was improved in all investigated cases. Using the PAMoCo algorithm we were able to reconstruct images from phases featuring strong motion, with an image

quality comparable to the one of the best phase reconstructions. Comparing the PAMoCo results to the MAM reconstructions, we see that the PAMoCo algorithm provides an

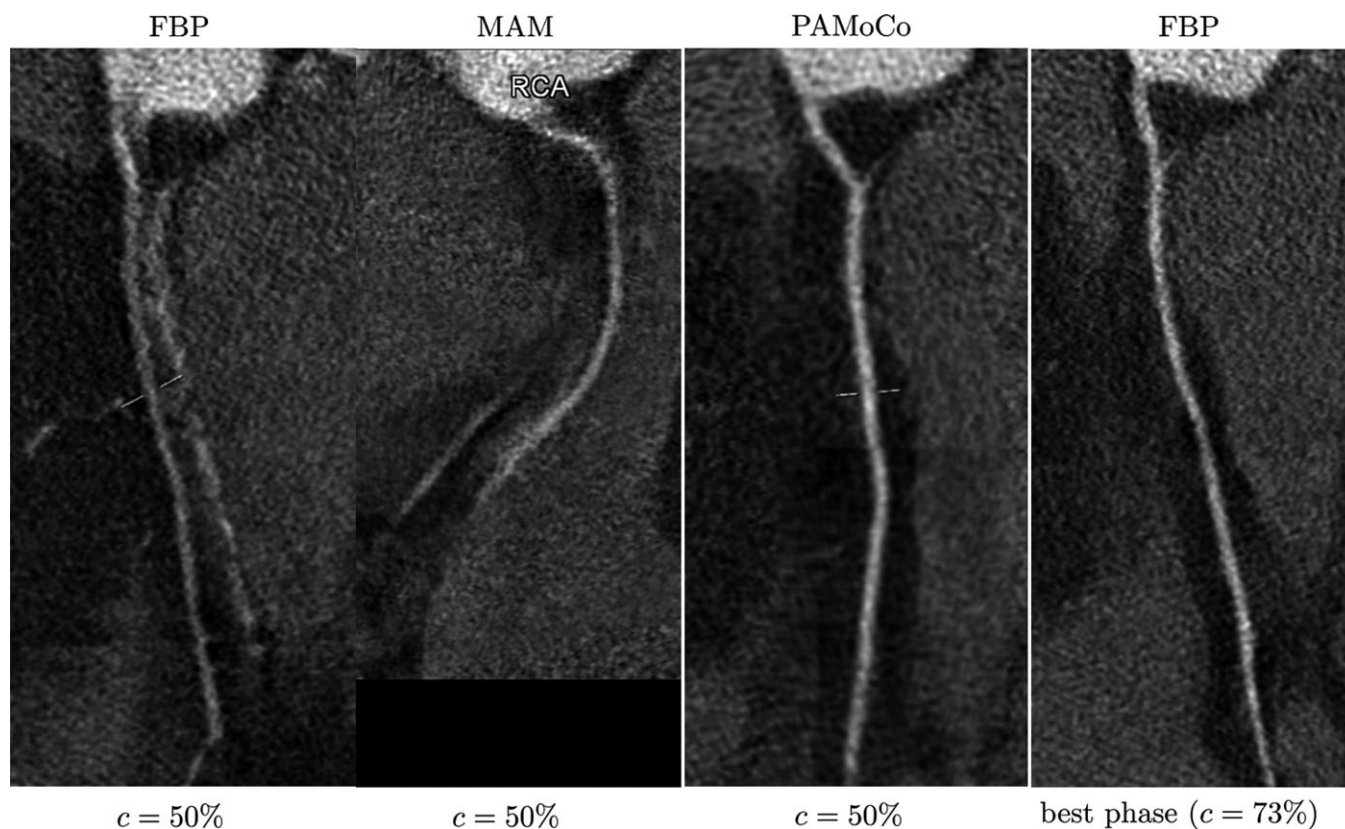


FIG. 19. Patient 2: The reduction of the motion induced doubling can as well be observed in the curved MPR derived from the PAMoCo reconstruction (third image), when compared with the FBP case (first image).

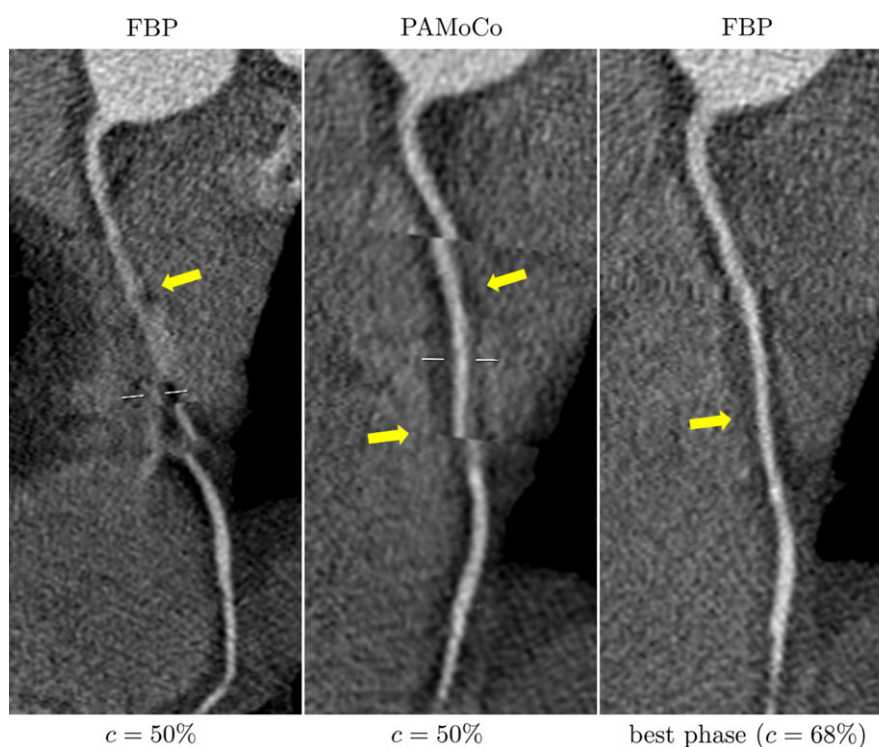


FIG. 20. Patient 3: A reduction of motion artifacts in the curved MPR derived from the PAMoCo reconstruction (third images) can especially be observed in the mid stack segment as emphasized by the arrows. Note furthermore that in this case in the PAMoCo reconstruction discontinuities occurring at the stack borders are clearly visible. [Color figure can be viewed at wileyonlinelibrary.com]

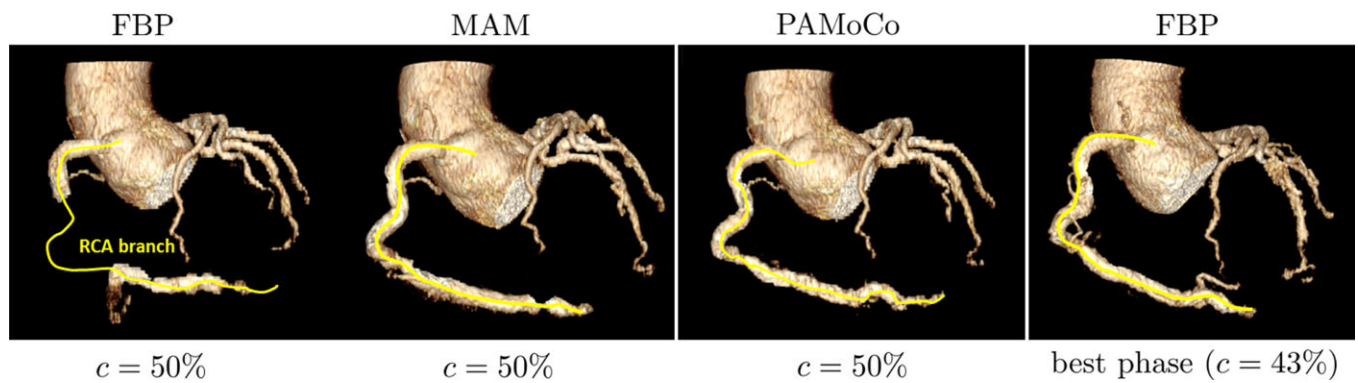


FIG. 21. Patient 1: Volume renderings of the coronary tree derived from FBP (first from left) and PAMoCo reconstructions (third from left) created with Syn-go[®]. Due to motion artifacts contained in the FBP reconstruction, a considerable amount of the artery was not distinguished from the background and is missing in the left image. [Color figure can be viewed at wileyonlinelibrary.com] [Color figure can be viewed at wileyonlinelibrary.com]

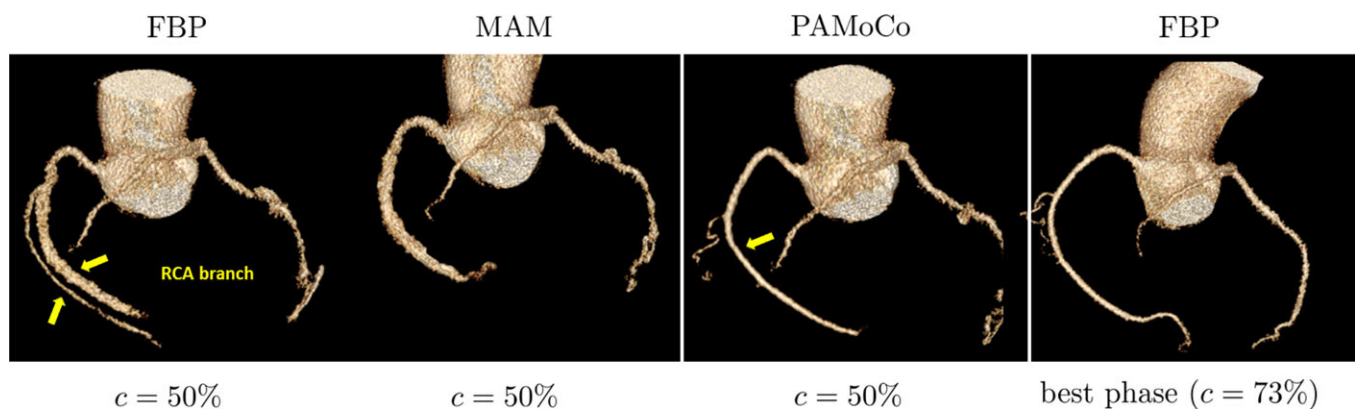


FIG. 22. Patient 2: The doubling of the RCA visible in the volume renderings of the coronary tree derived from the FBP reconstruction (first column) completely disappears when using PAMoCo (third column). [Color figure can be viewed at wileyonlinelibrary.com] [Color figure can be viewed at wileyonlinelibrary.com]

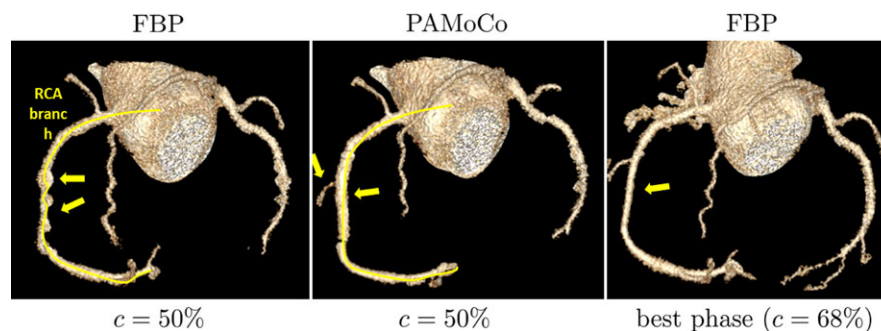


FIG. 23. Patient 3: The RCA visible in the volume renderings of the coronary tree derived from the PAMoCo reconstruction (3rd image) appears much smoother and even a small branch could be identified. [Color figure can be viewed at wileyonlinelibrary.com] [Color figure can be viewed at wileyonlinelibrary.com]

improved potential to reduce motion artifacts, since it performed better in all investigated cases.

4. DISCUSSION

The settings of the PAMoCo algorithm, as they have been proposed in this article, have been the same in all considered cases — both the phantom measurement and the clinical data — which is why we assume that the algorithm works without further tuning of parameters or user interaction.

The design of the PAMoCo algorithm is based on the ability to perform fast motion estimation. All components — the compensation of motion based on PARs, the way motion is modeled and the setup of the motion estimation routine comprising the reinitialization of the optimization — are constructed to allow fast PAMoCo, which should be as accurate as possible to provide images of sufficient quality to be diagnostic. The major source of computational costs originates from applying the MVF, which has to be performed already at least 1000 times during the brute force search, as described

in Section II.A.6. Applying the derived MVFs on the PARs it is possible to obtain MoCo reconstructions within a reasonable time window and hence to potentially integrate this method into clinical routine. In our implementation, which uses GPU programming to accelerate the calculation, the computation of a single image $f_{\text{MoCo}}(\mathbf{p})$ takes between 0.1–0.3 s. In contrast, a standard FBP reconstruction based on rebinned data, not providing motion-compensated reconstruction, of a volume of the same size already takes 8 s. In total, the computation of the PAMoCo algorithm for a single coronary artery takes between 10 and 15 min, which is comparable to the computation time of the MAM algorithm, obtaining motion-compensated reconstruction by back-projection along curved trajectories. The MAM images obey a lower image quality than the PAMoCo reconstruction concerning motion artifacts. However, note that due to the low-pass filtering characteristic of the forward projection and interpolation effects when adding the warped PARs, a slight smoothing effect appears inside the ROI in the PAMoCo reconstructions, which is not visible in the MAM images.

Based on the results of the phantom measurement we want to emphasize that PAMoCo is especially able to reduce motion artifacts significantly also in cardiac phases leading to strong motion artifacts, which are cases in which the MAM method most of the times broke down, as already emphasized by Rohkohl et al.¹⁷ This is a consequence of the special parameterization of motion introduced in Section II.A.4 combined with the fast way motion compensation is achieved, allowing for reinitialization of the optimization routine, as has been argued by Hahn et al.⁴ To document this ability, we applied PAMoCo to clinical data showing strong motion artifacts and were able to improve the delineation of the coronary arteries strongly as has been shown in the clinical cases in Section III.

From the phantom measurement we deduced that PAMoCo was also able to reduce motion artifacts significantly also in cardiac phases leading to strong motion artifacts. Hence, the image quality became less dependent on the heart rate, as the image quality increased throughout all simulated phases in case of both heart rates. The quality of the restoration of objects properties is also reflected by the maximum vessel diameter measurement. However, in some phases featuring large motion, for example, the phases between $c = 36\%$ and 60% for the 70 bpm measurement and $c = 20\%$ and 28% and 64% and 80% in case of the 90 bpm measurement, small motion artifacts still remained, which is also in agreement with the volume and maximum diameter measurements. The remaining of slight artifacts can in general be caused in case motion is not well described by the proposed model, or if the optimization routine does not get close enough to the global maximum. However, concerning our phantom measurement, where a smooth, continuous motion was applied, the motion model complex and the optimization routine was adequate enough to allow for a good estimation. Instead, the remaining of slight artifacts can be caused by inconsistencies appearing in the first and last PARs, induced by reconstruction from synthetic data, as

argued in Section II.A.2. However, in clinical cases, the appearing streak bands, when creating PARs from raw data, have a stronger influence, since they misguide the optimization process.

In general, the MAM and PAMoCo method depend on the presence of contrast media, which is necessary for the segmentation, but as well for the optimization of the sharpness measuring cost function. In cases of insufficient contrast, or high image noise, vessels of very small diameter and the presence of severe motion artifacts, segmentation might partially fail and prevent motion compensation. Although, since the cost function is sensitive to high-contrast objects, we suggest that this method can especially be recommended, for example, the assessment of stent patency in single-source cardiac CT or calcium scoring based on contrast-enhanced scans. In addition, FFR CT measurements are likely to benefit from MoCo as well.

Furthermore, the algorithm can be extended to other application like contrast-enhanced high-pitch dual-source spiral scans of the thorax region, which might then be used for cardiac examinations as well.

5. CONCLUSIONS

We proposed an algorithm for MoCo in the region of the coronary arteries, which is designed to be applied on data acquired in dose-optimized scan protocols by utilizing only the data needed for the reconstruction of a single heart phase. The algorithm is recommended to be applied to standard cardiac CT scans acquired with SSCT systems, such as sequence scans or low-pitch spiral scans using tube current modulation.

The MVF-determining parameters were estimated optimizing an image artifact measuring cost function, as proposed by Rohkohl et al.¹⁷ By introducing a suitable parameterization we were able to limit the number of independent parameters and at the same time regularize the MVF. The determined MVFs were applied to PARs allowing for fast motion-compensated reconstruction. With this reconstruction and the choice of parameterization, we were able to design a pseudo-global optimization routine to overcome the issue of getting trapped in a local minimum of the nonconvex cost function. In this way we were able to increase the potential of the previous MoCo algorithm, from cardiac phases featuring only small motion artifacts, to a wider range of initial image quality considering the influence of motion.

Based on a phantom measurement we were able to show that the image quality of PAMoCo reconstructions becomes almost independent of the chosen reconstruction phase. We verified our findings in six clinical cases, where we observed that the image quality of the original FBP reconstruction was remarkably increased when using PAMoCo, since typical motion artifacts, such as a blurring of the coronary arteries or shading around the latter, were diminished.

In future research it might be worthwhile to further extend the cost function to achieve higher sensitivity to motion artifacts or to think of a more sophisticated optimization routine.

In addition, the problem of discontinuities appearing at the borders of the stacks, as explained in Section I, will be addressed.

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CONFLICTS OF INTEREST

The authors have no relevant conflicts of interest to disclose with respect to the research, authorship, and/or publication of this paper.

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